



Industrial Oriented Mini Project Registration Form (AY 2025-26)

1. Date of Registration: 30-12-2025
2. Students' details:

3. Title of the Project: Fullstack with ML integrated project.
4. Name of the Supervisor: Mr.Suhail Afroz
5. Signature of the students: i. ii.
6. Signature of the Supervisor:
7. Name & Signature of Section Coordinator: Ms. Duba Sriveni

Dr. A. Vani Vathsala
Professor & Head, Dept of CSE

INDUSTRIAL ORIENTED MINI PROJECT (IOMP)

ABSTRACT :

This paper presents an advanced full-stack software system that seamlessly integrates machine learning, interactive visualization, intelligent recommendation modeling, and a conversational AI assistant into a unified application architecture, with the primary objective of delivering a highly scalable and data-driven platform capable of supporting dynamic user interactions, personalized insights, and automated decision-making. The solution follows a modular multi-tier architecture consisting of a modern frontend interface, a robust backend service layer, and a dedicated machine learning microservice module, where the backend manages core functionalities such as authentication, API orchestration, data handling, and inter-component communication, while the Python-based ML microservices perform predictive analytics, clustering, pattern recognition, and personalized recommendations through structured model pipelines. To enhance usability and interpretability, the system incorporates an interactive visualization component that transforms raw datasets into intuitive dashboards for exploring trends, anomalies, and insights, along with an NLP-driven chatbot that enables natural language interaction, automated query handling, and context-aware system assistance. The integration of full-stack development, machine learning intelligence, and interactive visualization results in a comprehensive and adaptable platform suitable for diverse real-world applications, demonstrating how AI-powered features can be effectively embedded into modern web architectures to improve user engagement, decision support, and operational efficiency.